

COURSE DESCRIPTION

Course Number	0418-126	Course Title	Computer Programming I
Credit Hours	3 (3:1:3)	Course Coordinator	Dr. Hussain Almohri
Compulsory	No	Prerequisite	None

Textbooks

David I. Schneider (2016), Introduction to Programming Using Python plus MyProgrammingLab, Pearson.

Robert Sedgewick, Kevin Wayne, Robert Dondero (2015), Introduction to Programming in Python: An Interdisciplinary Approach, Addison-Wesley Professional.

Course Hours and Topics

	Topic	Hours
1	Introduction to basics and history of computers, machine structure, programming languages, and the Python programming language	3
2	Interactive computing (Python shell), Python as a calculator	2
3	Expressions, reserved words, variables, numeric data types, comments, libraries	3
4	Boolean expressions, conditionals, loops	6
5	Function definition, function calls, parameter passing, return values	6
6	File creation and manipulation	3
7	Strings, lists, enumeration, arrays, dictionaries, sets, pointers (concept)	4
8	Python libraries, date and time, random numbers, mathematics and statistics functions, system functions, exceptions	3
9	Simple search and sort algorithms	3
10	Programming scientific and educational applications	6
11	Exams	3
	Total	42

Lab Hours

	Topic
1	Python installation and Python IDE (PyCharm)
2	The use of command line terminal, interactive computing with Python shell
3	Expressions and datatypes
4	Boolean data type and conditionals
5	Loops (for and while)
6	Functions
7	Return values
8	Text file creation, read, and write
9	Working with data and CSV files
10	Strings, arrays, and lists
11	Dictionaries, enumeration, sets
12	Python libraries and exceptions
13	Python applications in science
14	Python applications in education

Grading

15–20%	Midterm exam I
15–20%	Midterm exam II
20–30%	Homework assignments, lab assignments, and quizzes
40%	Final

Course Outcomes

1	Students should understand the way computers work and the need for programming.
2	Students should learn to program using basic Python language constructs.
3	Students should understand the concepts of datatypes, conditionals, and loops.
4	Students should understand the concepts of programming libraries and modular programming.
5	Students should learn to use Python for building scientific and educational applications.